

Final Project Report

BUS-235F – Forecasting for Business Analytics

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Data Overview

The dataset contains the number of Uber pickups in New York City, categorized by pickup location, for a 6-month period between 01/01/2015 to 06/30/2015. The data is sampled at 1 hour rate and contains 29102 observations. The dataset also has information for corresponding weather conditions such as wind speed, visibility, temperature, precipitation, and snow depth; as well as holiday information; which may be possible factors that affect the number of pickups. Overall, the dataset has 2 non-numeric (category) variables and 10 numeric variables.

Dataset Reference:

Yannis Pappas. Jan, 2016. NYC Uber Pickups with Weather and Holidays, 1. Retrieved 08/30/2024 from <https://www.kaggle.com/datasets/yannisp/uber-pickups-enriched>

Note - The dataset used for this project is a subset of the larger dataset that was obtained by [FiveThirtyEight](#) from the [NYC Taxi and Limousine Commission](#) by submitting a Freedom of Information Law request on July 20, 2015. The dataset creator referenced above has used a subset of this data and enriched it with weather information from [National Centers for Environmental Information](#).

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Exploratory Data Analysis

In this dataset, data is not segregated by boroughs resulting in time duplicates in the pickup_dt column. To create a regular time series for each borough we filter by borough and then set pickup_dt as the index. After processing the data in this way we can plot the pickups over time for each borough. The pickup data for Brooklyn, Manhattan and Bronx is shown in figure 1.

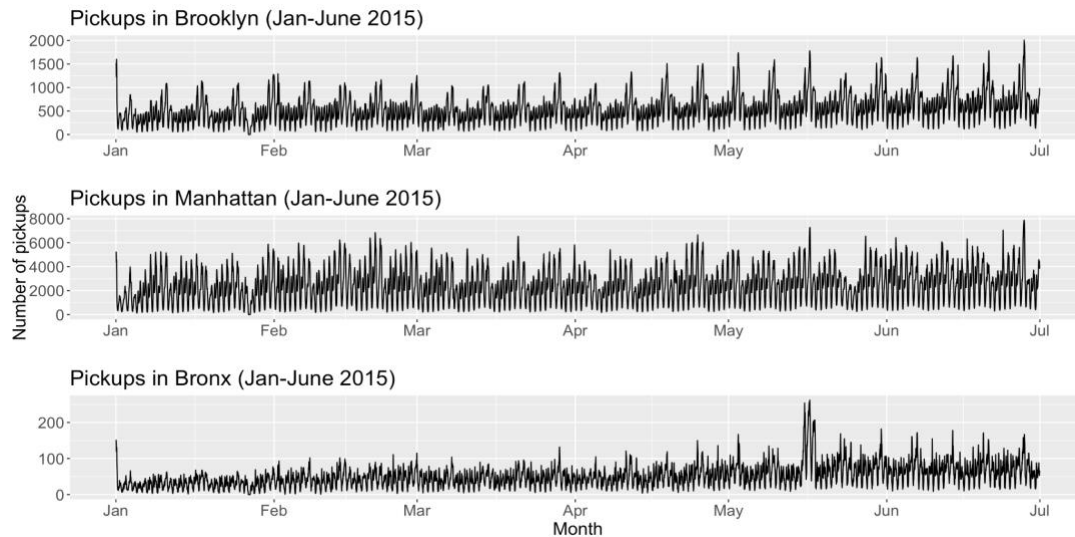


Figure 1: Visualization of the number of Uber pickups in three boroughs of New York City - Brooklyn, Manhattan and Bronx, from January 2015 to June 2015.

Since the data is quite dense, we can zoom into the last month of data (June) to identify patterns (Figure 2). We have annotated the day of the week on the x-axis to identify the day of the week.

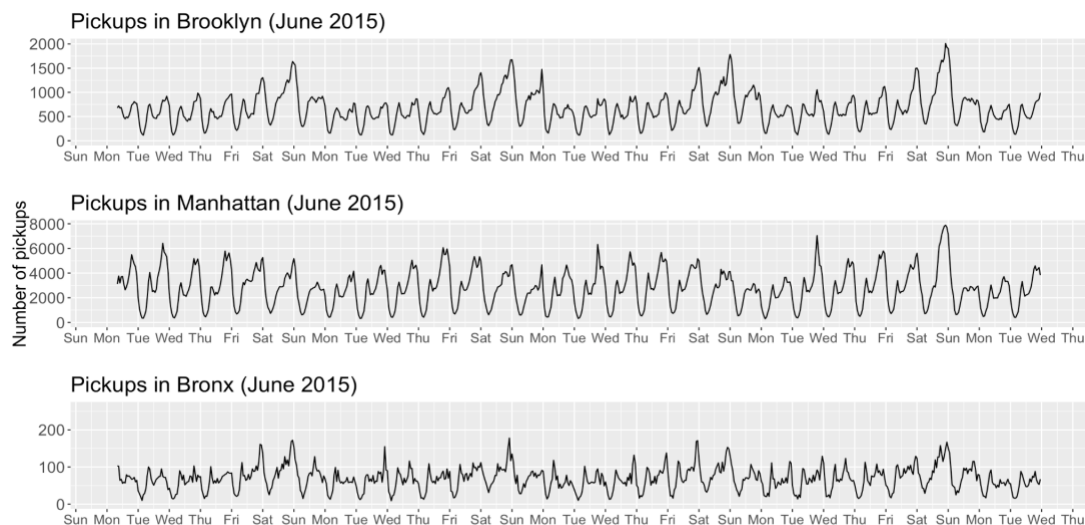


Figure 2: Visualization of the number of Uber pickups in three boroughs of New York City - Brooklyn, Manhattan and Bronx, for the month of June 2015. This zoomed-in view highlights the seasonality patterns in the dataset.

As shown in Figure 1, pickups in every borough show a consistent upward trend over time. Furthermore, we observe in Figure 2, that there are two levels of seasonality in the data at daily and weekly levels. We note that the number of pickups is generally higher on Friday - Sunday compared to the other days of the

week. We also note that pickups within a particular day show a bimodal distribution with higher number of pickups before noon and in the evening/night.

As described earlier, our dataset contains information regarding weather conditions alongside pickup data. The data includes columns for wind speed, visibility, temperature, dew point, sea level pressure, 1-hour, 6-hour and 24-hour precipitation, and snow depth. With the goal of identifying predictors that could aid our forecasting for pickups, we calculated correlations between the number of pickups and each weather condition variable detailed above (listed in Table 1). We note that only 2 predictors - temperature and dew point show a moderate correlation with the number of pickups. We also notice a weak negative correlation with 24-hour precipitation and snow depth, which is expected since people are less likely to travel during rain/snowfall. The data showing these moderate correlations is plotted in Figure 2. As we do not observe a strong correlation of the number of pickups on weather variables, we conclude that a linear regression model using the weather variables as predictors would not be feasible.

Weather variable	Correlation with number of pickups
wind speed	-0.044
visibility	-0.004
temperature	0.311
dew point	0.239
sea level pressure	-0.053
1-hour precipitation	0.018
6-hour precipitation	0.024
24-hour precipitation	-0.102
snow depth	-0.135

Table 1: Correlation metrics for the various predictors within the dataset. We observe small correlations of pickups with temperature, precipitation and snow depth.

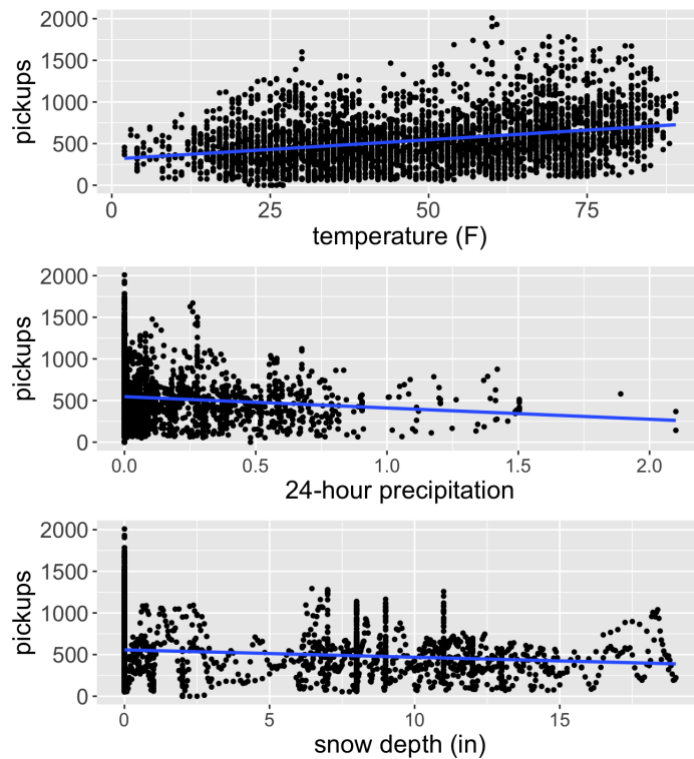


Figure 3: Scatter plots depicting correlations between a select few predictors within the dataset that are moderately correlated.

In the next section, we will use a variety of forecasting techniques to identify the model best suited for predicting the number of pickups in various boroughs. For this purpose, we have split the data as 80 percent for training and 20 percent for testing. Hence, training data for each borough spans the months of January to May and test data is for the month of June.

Forecasting

Benchmarking Forecasting Methods

We have applied Mean, Naive, Seasonal naive and Drift benchmark models to each borough's pickup data. Figure 4 shows the predictions of the Mean, Naive and Drift models for each borough.

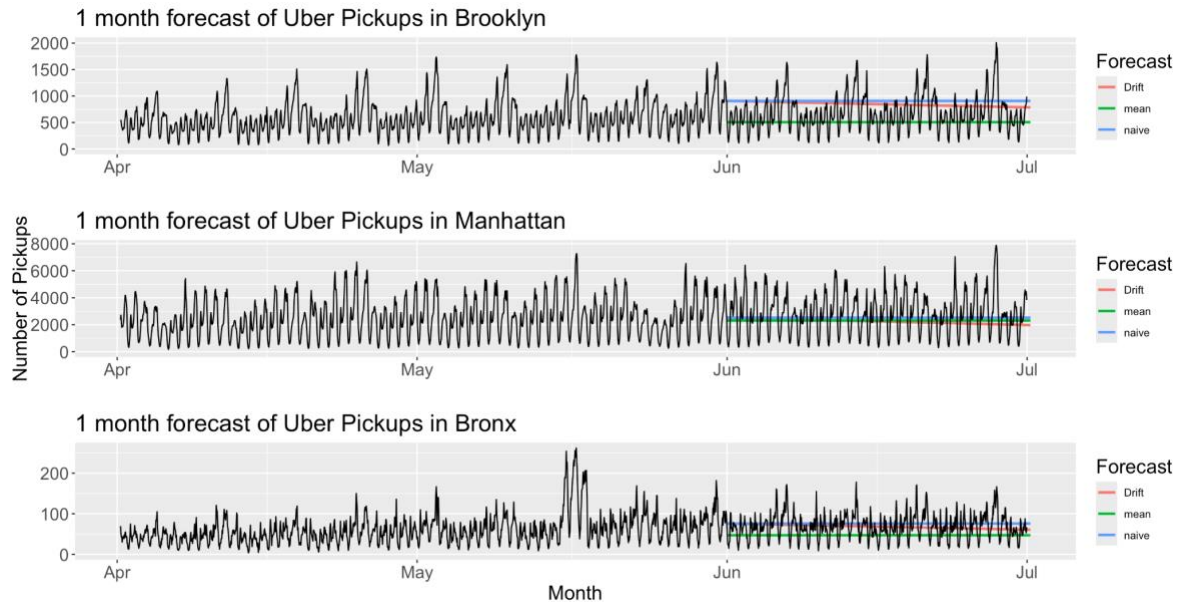


Figure 4: Predictions for benchmark forecasting methods - Mean, Naive and Drift for three New York boroughs.

The plots for the Seasonal Naive model are shown separately in Figure 5 for clarity.

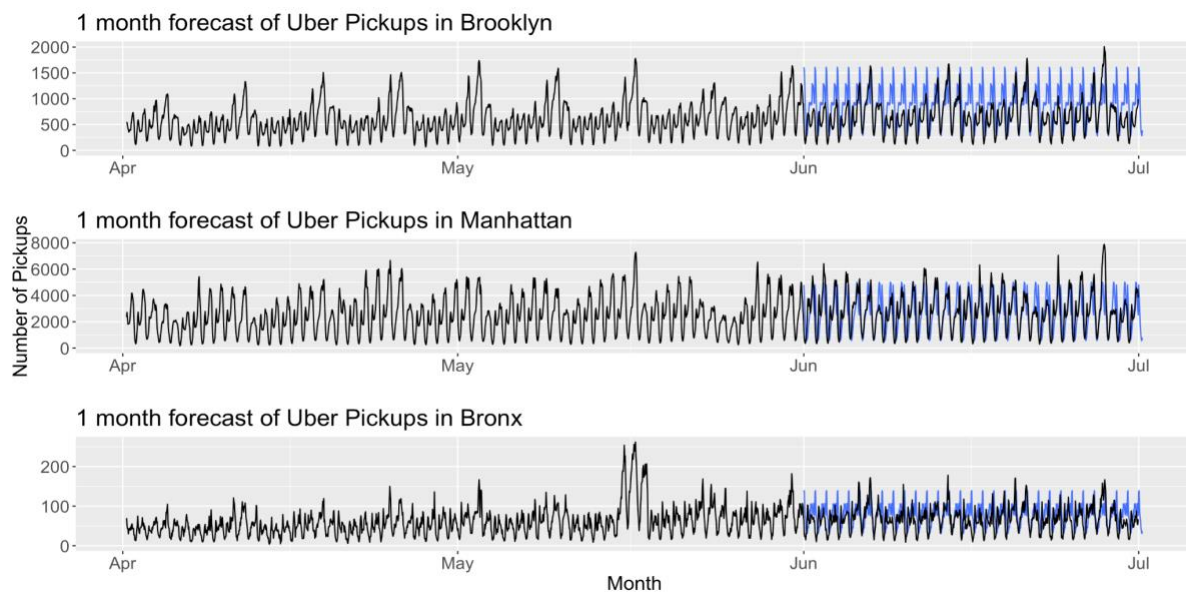


Figure 5: Prediction for benchmark forecasting method - Seasonal Naive for three New York boroughs.

The accuracies for these models compared against the test dataset is shown in Table 2. Interestingly, a different model works best for each borough. Since the data is highly seasonal, we were expecting the seasonal naive model to outperform others for all boroughs, however, this is not the case. The seasonal naive model replicates the data from the last day of the month which happens to be a weekend. Since the number of pickups is typically higher on weekends, this biases the forecast to be on the high end for most days of the week in the test dataset. Hence we get larger errors on seasonal naive compared to other benchmark models for Bronx and Brooklyn. For Manhattan, the gap between the number of pickups on weekends vs weekdays is smaller, hence the seasonal naive model performs better.

	Brooklyn			Manhattan			Bronx		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
Mean	367	269	47.6	1562	1244	76.9	37.4	29.8	49.8
Naive	406	342	91.9	1513	1196	81.8	32.0	25.1	66.2
Snaive	443	338	82.6	1275	933	56.3	38.1	29.5	67.8
Drift	379	310	82.6	1592	1271	76.5	31.5	24.2	58.7

Table 2: Accuracy metrics for benchmark forecasting methods

Exponential Smoothing Models

In this section, we explore the forecasting performance of exponential smoothing models on our dataset. Since the pickup data for each borough show an upward trend over time and seasonality, we have chosen the Winter's Method as our method-of-choice, as it can account for both. We have trained three variations of Winter's methods - Additive, Multiplicative, and Damped. The accuracy of each of these models on the test dataset is tabulated below in Table 3.

	Brooklyn			Manhattan			Bronx		
Winter's	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
Additive	2442	2107	395	2220	2018	96.8	24.0	17.9	34.2
Multiplicative	5009	4016	634	25209	19348	742	231	186	287
Damped	274	224	39.5	967	692	27.1	24.2	18.4	33.4

Table 3: Accuracy metrics for exponential smoothing models.

We can see that the damped version of Winter's method works best for Brooklyn and Manhattan by a large margin. For Bronx, we find that Winter's additive method wins over the damped method, but only narrowly. The forecasts for Winters multiplicative method seem to diverge and produce larger forecast errors in all cases.

The forecast plots, along with fitted values for the best exponential smoothing model for each borough is shown in Figure 6.

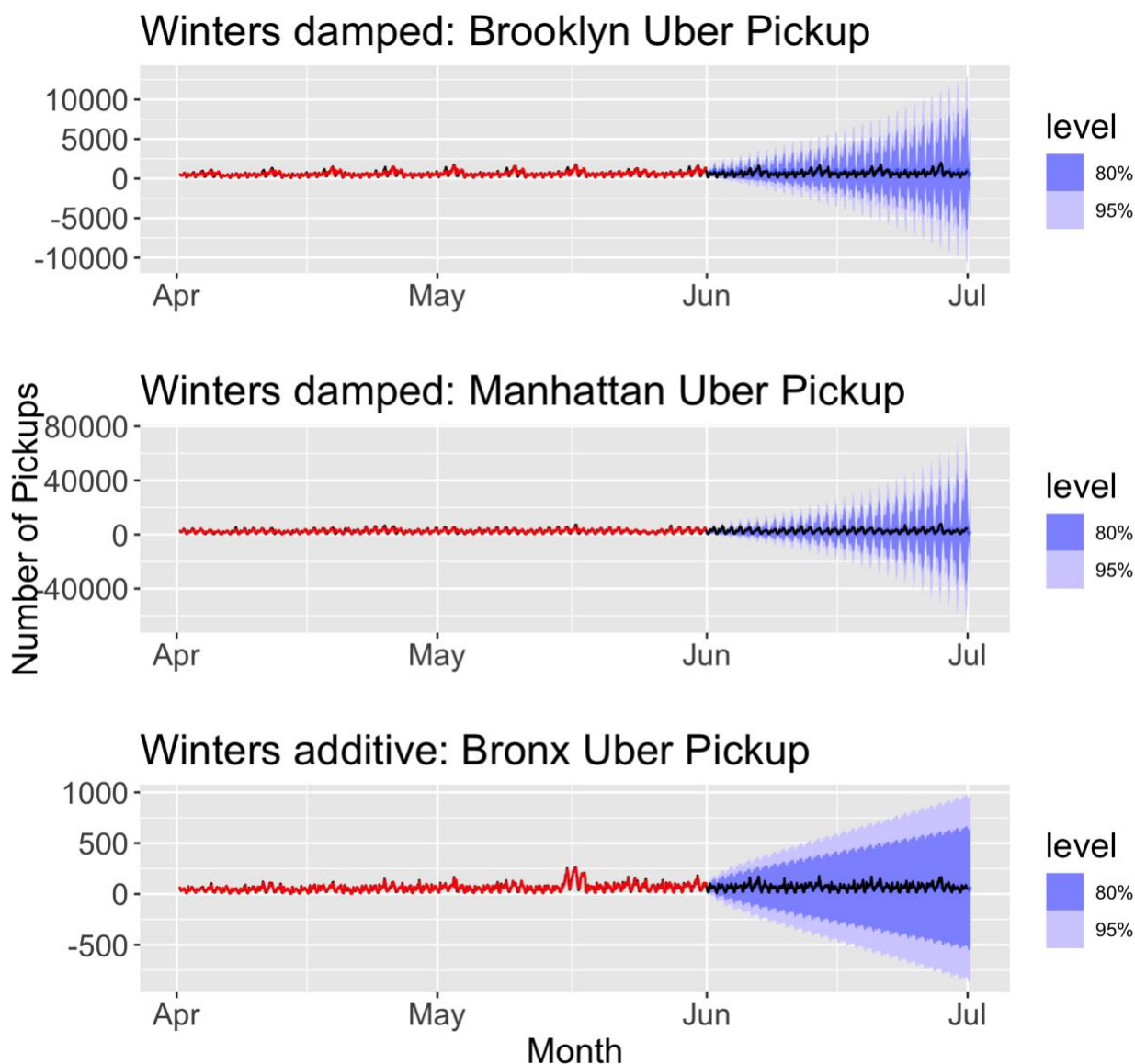


Figure 6: Predictions for the highest accuracy exponential smoothing method for three New York boroughs along with prediction intervals and fitted values (in red).

We can see that although the accuracy of these models on the test set is better than the benchmark models, the prediction interval over a 1-month forecast horizon increases significantly. Next, we will fit Harmonic regression models to these datasets.

Harmonic Regression Models

In this section, we explore the forecasting performance of harmonic regression models on our dataset. Harmonic regression models should be able to account for multiple levels of seasonality in the data which can be essentially represented by different harmonics. We have trained six harmonic regression models for

values of K ranging from 1 to 6. The accuracy of each of these models on the test dataset is tabulated below in Table 4.

	Brooklyn			Manhattan			Bronx		
Harmonic Regression	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
K = 1	289	225	49.2	1056	826	49.7	27.8	22.3	52.7
K = 2	257	187	36.6	886	659	34.3	25.8	21.0	47.5
K = 3	252	182	34.8	873	647	33.2	24.7	20.0	44.5
K = 4	252	182	34.6	874	645	33.0	24.8	19.9	44.6
K = 5	251	180	34.5	860	634	33.0	24.5	19.5	44.3
K = 6	251	180	34.5	860	633	33.1	24.5	19.5	44.2

Table 4: Accuracy metrics for harmonic regression models.

We can see that models with K=5 and K=6 provide the best (and nearly equal) accuracy on the test dataset amongst all harmonic regression models. To visualize how well these models predict the number of pickups, we have plotted the fitted values on the training set and forecasted values against the test set in Figure 7.



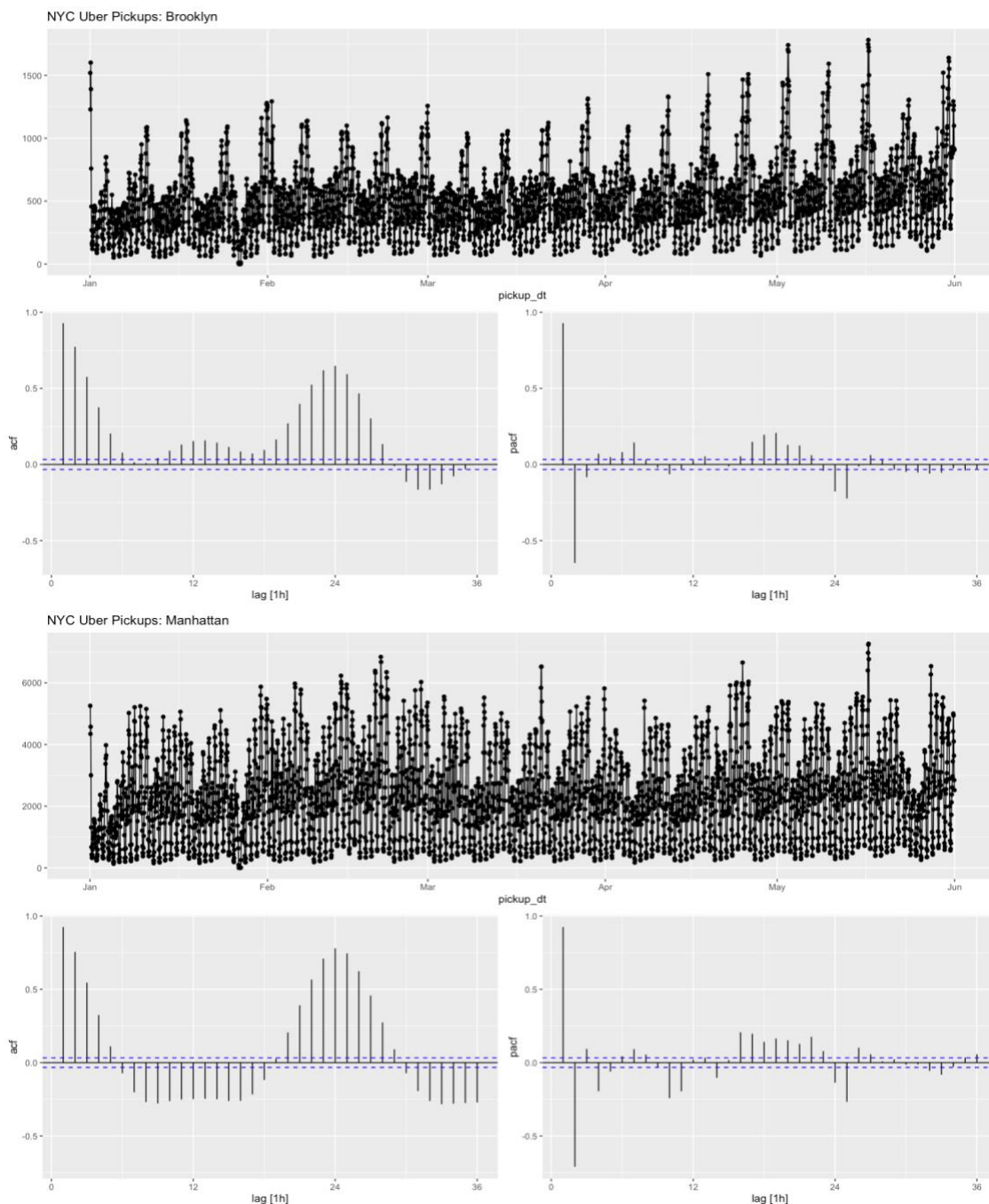
Figure 7: Predictions for the highest accuracy harmonic regression method for three New York boroughs along with prediction intervals and fitted values (in red).

We note that the Harmonic regression models have captured the daily seasonality in the data very well. It is able to reproduce the bimodal distribution of pickups over the day with peaks during rush hours.

However, it has not been able to capture the weekly seasonality well, i.e. higher number of pickups on the weekends. Increasing the value of K to 12 had no impact on the accuracy of the model over the test dataset. It is also worth noting that the prediction intervals are much narrower compared to exponential smoothing models that were discussed in the previous section.

ARIMA Models

In this section, we will explore the ARIMA model for forecasting pickups at 3 boroughs - Brooklyn, Manhattan and Bronx. As discussed in previous sections, pickup data has both seasonality and trend and therefore time series is not stationary. We verified it by time plot, ACF and calculating p value (Figure 8).



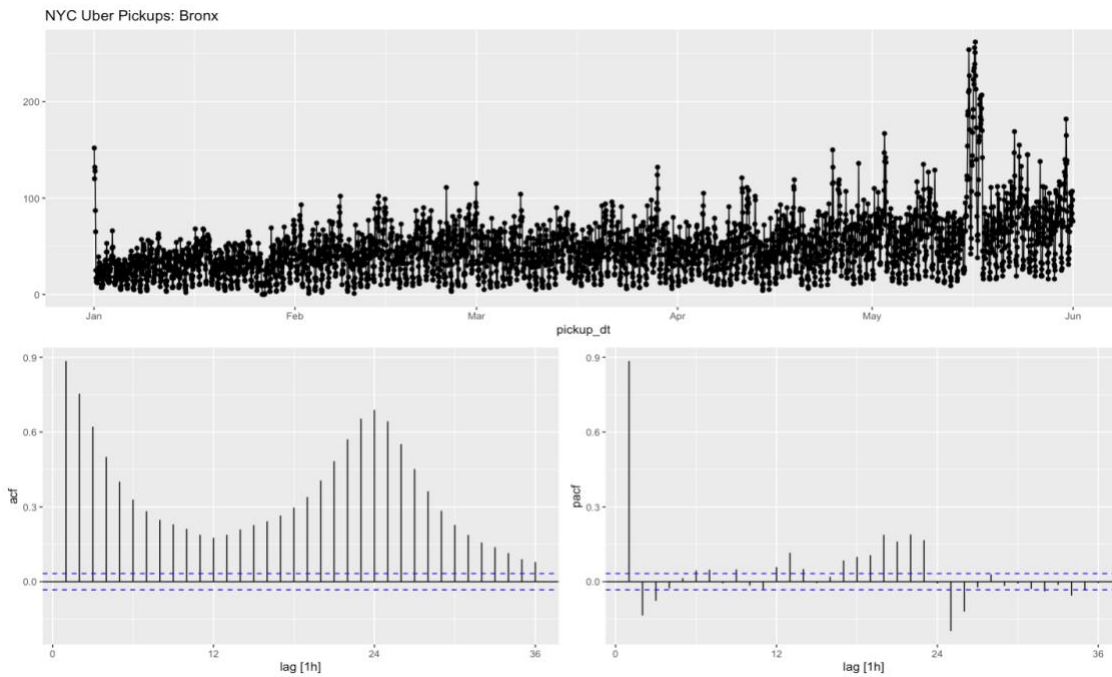
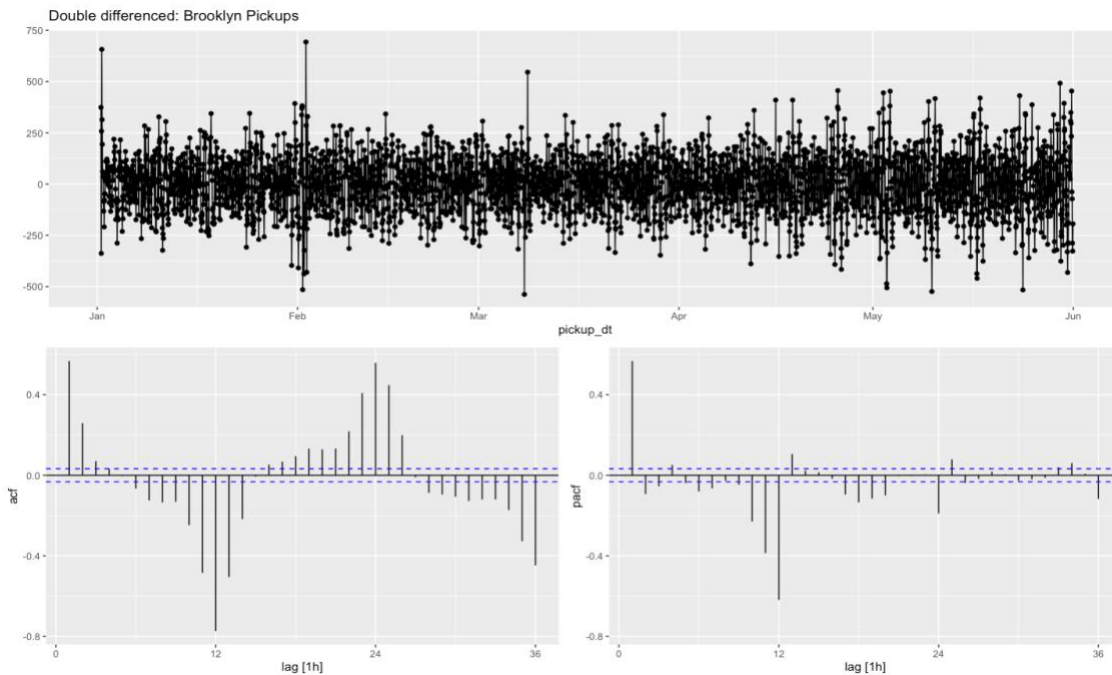


Figure 8: Time series data along with its ACF and PACF for three New York boroughs.

Looking at the time plot, ACF value does not drop quickly for all the boroughs and for the kpss test, they have a p value of 0.01. Since it has a small p of 0.01, it indicates that the null hypothesis can be rejected and the data is not stationary. Then we looked at the seasonal difference and first difference for all three boroughs and applied an appropriate differencing method to make the time series stationary. We also checked the p-value for the kpss test after applying differencing and got a p-value of 0.1. We can conclude that the differenced data is stationary (Figure 9).



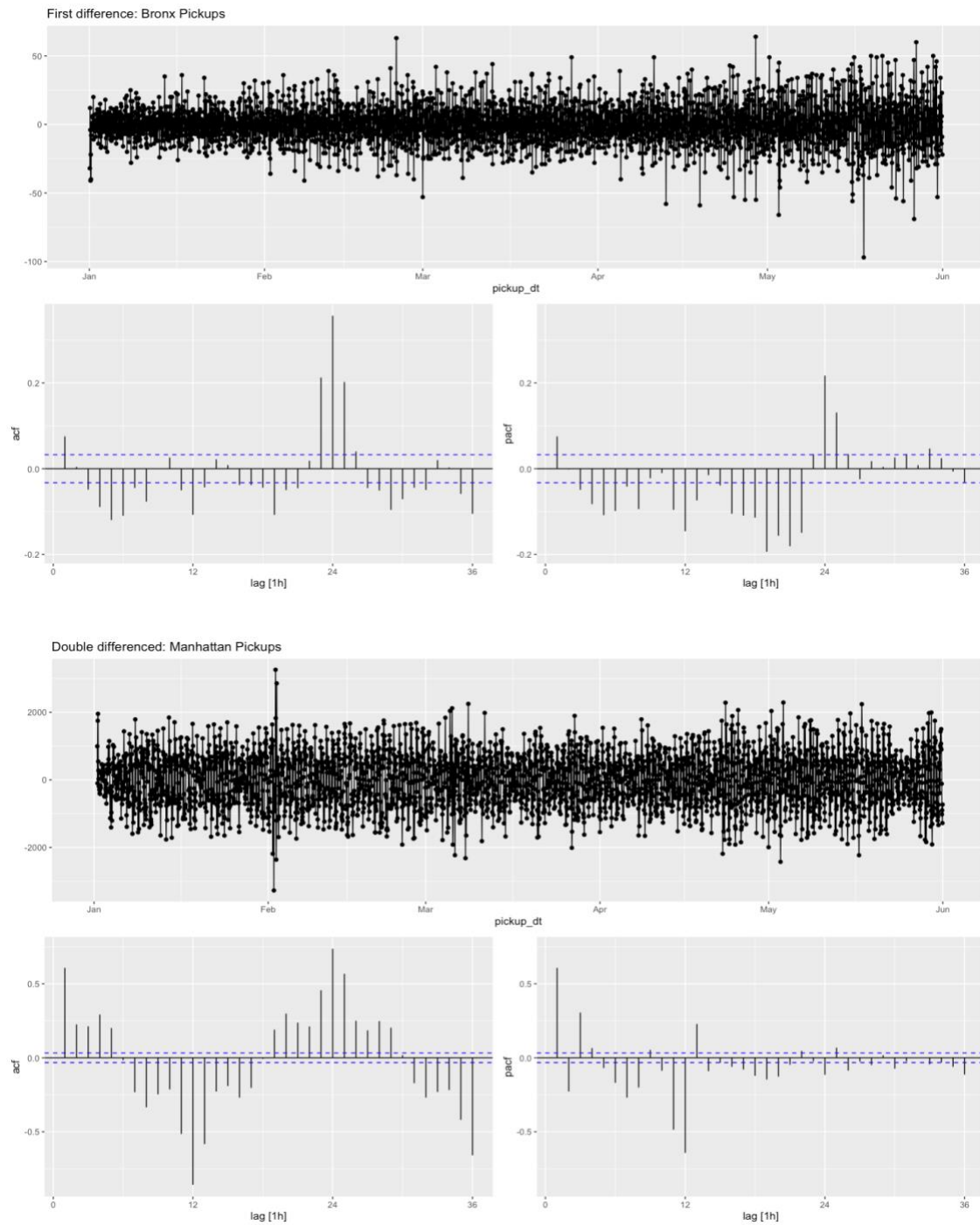


Figure 9: Time series data along with its ACF and PACF after differencing for three New York boroughs.

Looking at the ACF and PACF graph in Figure 9, parameters (pdq) values for Non-seasonal part and (PDQ) values for seasonal part have been identified and appropriate ARIMA models were applied to fit the training data set. Residual analysis was performed to check the accuracy of the model for the test set. The accuracy of each of these models on the test dataset is tabulated below in Table 5 .

	Brooklyn			Manhattan			Bronx		
ARIMA	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA113111	432	349	53.6	N/A	N/A	N/A	N/A	N/A	N/A
ARIMA311111	308	273	63.1	N/A	N/A	N/A	N/A	N/A	N/A
ARIMA013012	N/A	N/A	N/A	2932	2779	143	39.7	32.5	50.1
ARIMA310200	N/A	N/A	N/A	27067	30780	1554	57.2	48.2	64.0

Table 5: Accuracy metrics for ARIMA models.

To visualize how well these models predict the number of pickups, we have plotted the fitted values on the training set and forecasted values against the test set in Figure 10.

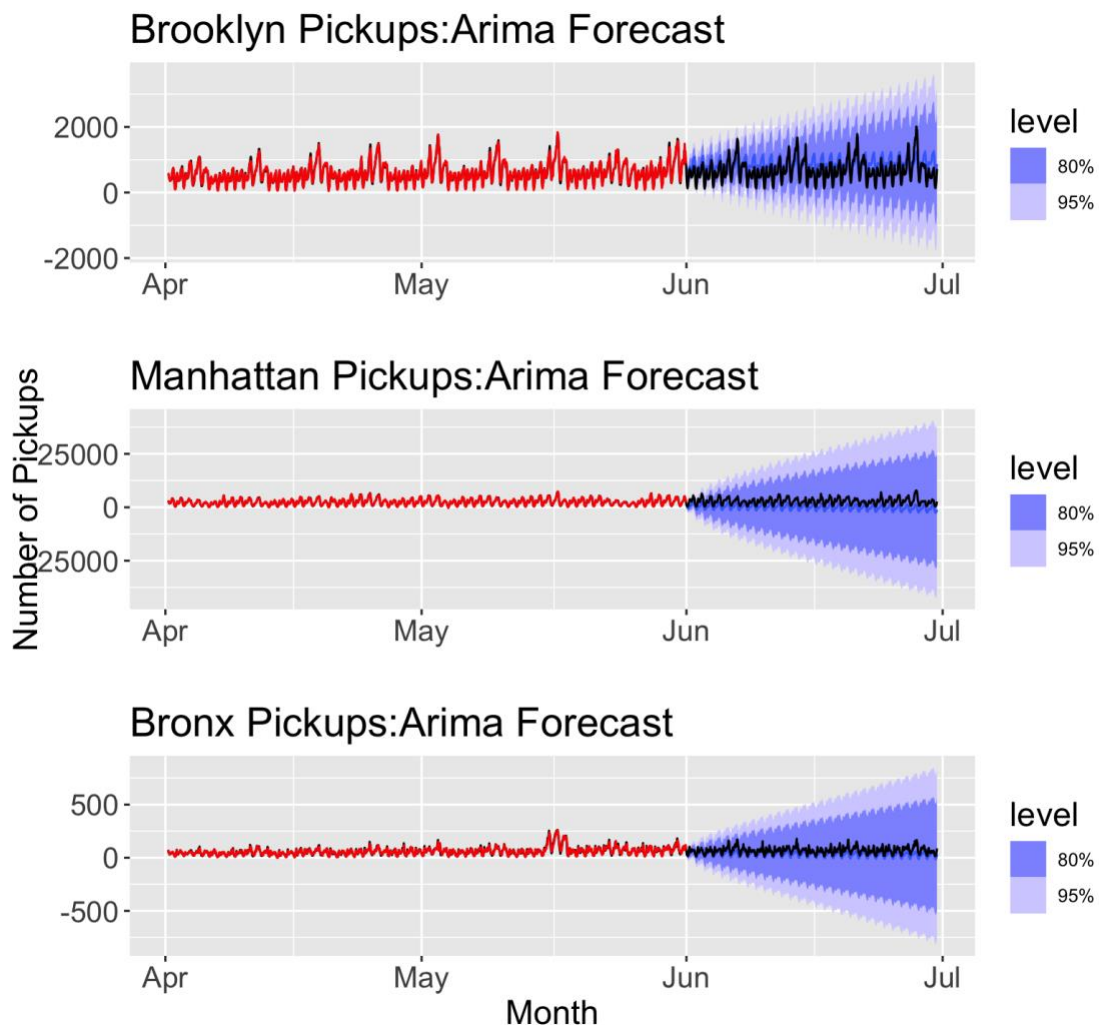


Figure 10: Predictions for the highest accuracy ARIMA model for three New York boroughs along with prediction intervals and fitted values (in red).

We note that the ARIMA model has captured the daily seasonality of the training data quite well. It was also able to judge the trend accurately for Brooklyn, but not for Bronx and Manhattan. These models were also not able to capture the weekly seasonality well, i.e. higher number of pickups on the weekends.

Neural Network Models

In this section, we explore the forecasting performance of neural network autoregression models on our dataset. Neural networks provide a powerful method to capture non-apparent patterns in the dataset. In this project, we have trained a neural network NNAR(p,k) model per borough to predict pickups.

The value for p was determined by looking at the PACF plot of the training data for each borough. Based on the last significant spikes, the value of p was chosen as 27, 25 and 26 for Brooklyn, Manhattan and Bronx respectively. The value of k was set to $\text{floor}((p+1)/2)$ yielding values 14, 13, 13 respectively. Training and forecasting a single neural network autoregressive model took more than 90 minutes to complete in R. As such, only one neural network model per borough was trained.

The accuracy of each of these models on the test dataset is tabulated below in Table 6.

Model	Brooklyn			Manhattan			Bronx		
Neural Network	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
p, k = 27, 14	362	256	37.6						
p, k = 25, 13				1331	1027	50.5			
p, k = 26, 13							32.7	25.1	45.7

Table 6: Accuracy metrics for neural network models.

To visualize how well these neural network-based models perform on the test dataset, the predictions have been plotted alongside fitted values in Figure 11 with prediction intervals.

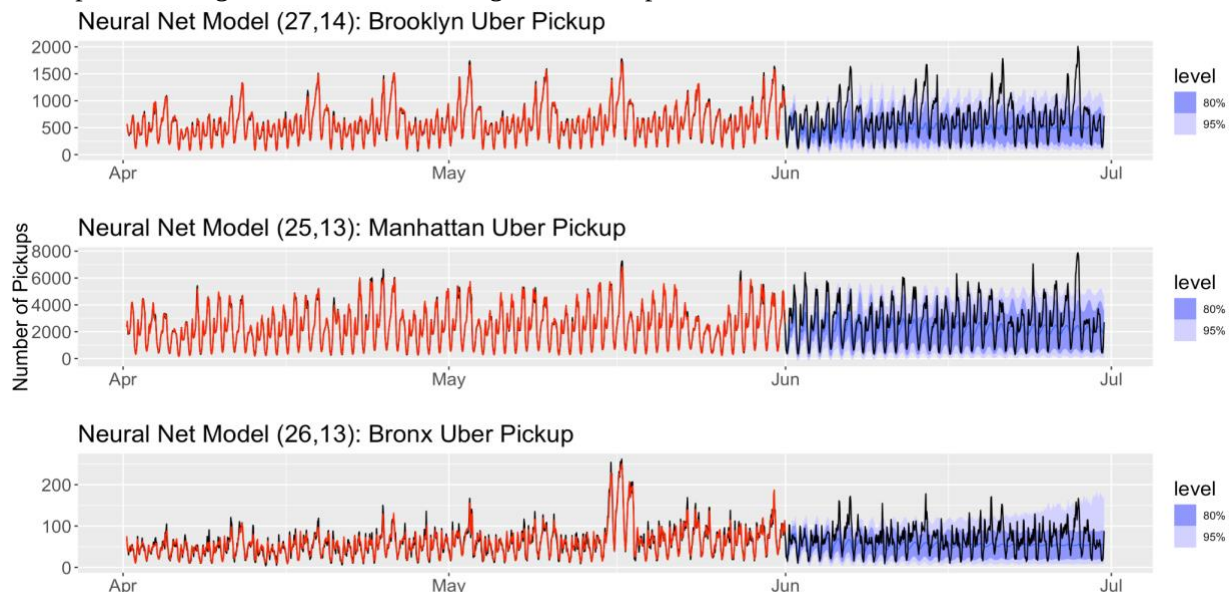


Figure 11: Predictions for the neural network method for three New York boroughs along with prediction intervals and fitted values (in red).

As the prediction intervals make it difficult to visualize the forecasts, they have been plotted without prediction intervals in Figure 12.

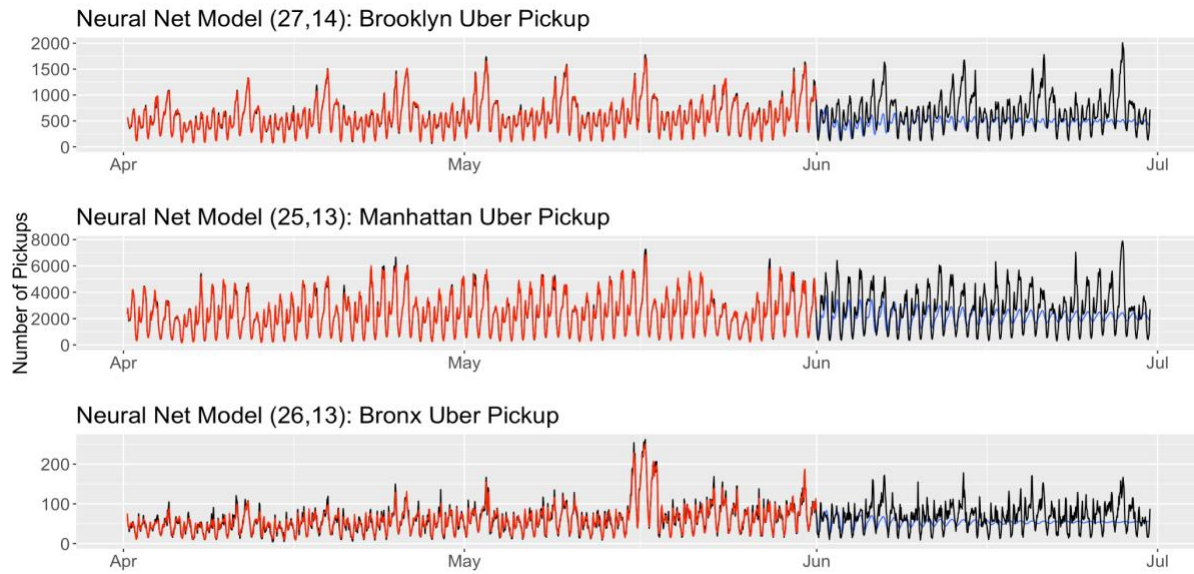


Figure 12: Predictions for the neural network method for three New York boroughs without prediction intervals (for easier visualization) and fitted values (in red).

We note that the predictions of the neural network models are quite accurate for the first 24 hours, but the accuracy drops off significantly as the prediction horizon increases. The magnitude of seasonal variation keeps dropping over time. This can be explained by the fact that we chose $p=24$, essentially only giving the neural network enough information regarding one cycle of data. Hence the model is not able to capture patterns which occur over a longer time scale than 24 hours.

Error and Model Recommendation

The final summary table comprising the best accuracy model of each type is shown in Table 7

Model	Brooklyn			Manhattan			Bronx		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
Mean	367	269	47.6	1562	1244	76.9	37.4	29.8	49.8
Naive	406	342	91.9	1513	1196	81.8	32.0	25.1	66.2
Seasonal Naive	443	338	82.6	1275	933	56.3	38.1	29.5	67.8
Drift	379	310	82.6	1592	1271	76.5	31.5	24.2	58.7
Exponential Smoothing	274	224	39.5	967	692	27.1	24.0	17.9	34.2
Harmonic Regression	251	180	34.5	860	634	33.0	24.5	19.5	44.3
ARIMA	308	273	63.1	2932	2779	143	39.7	32.5	50.1
Neural Network	362	256	37.6	1331	1027	50.5	32.7	25.1	45.7

Table 7: Summary of accuracy metrics from the best forecasting model of each type. Harmonic regression model is the better model for Brooklyn and Manhattan while Winter's additive model works best for Bronx.

As we can see, Harmonic regression model with K=5 is the better model for Brooklyn and Manhattan, while Winter's method - Additive is the better model for Bronx.

ARIMA models and neural network models do not fare as well for our dataset. The likely reason is that our dataset has hourly data, and we are predicting for a period of 1 month = 720 data points. ARIMA and neural network models are likely not very well-suited for this scenario since the size of the models would have to be prohibitively large to accurately predict over such long prediction horizons.

Insights and Conclusions

We have summarized the business insights generated through our analyses as responses to the questions first posed in project deliverable 1.

What times of the day is surge pricing most likely to occur due to high pickup requests?

We analyzed 3 of the boroughs in the New York area, Manhattan, Bronx, and Brooklyn. See the Weekly Patterns Figure 13 below:

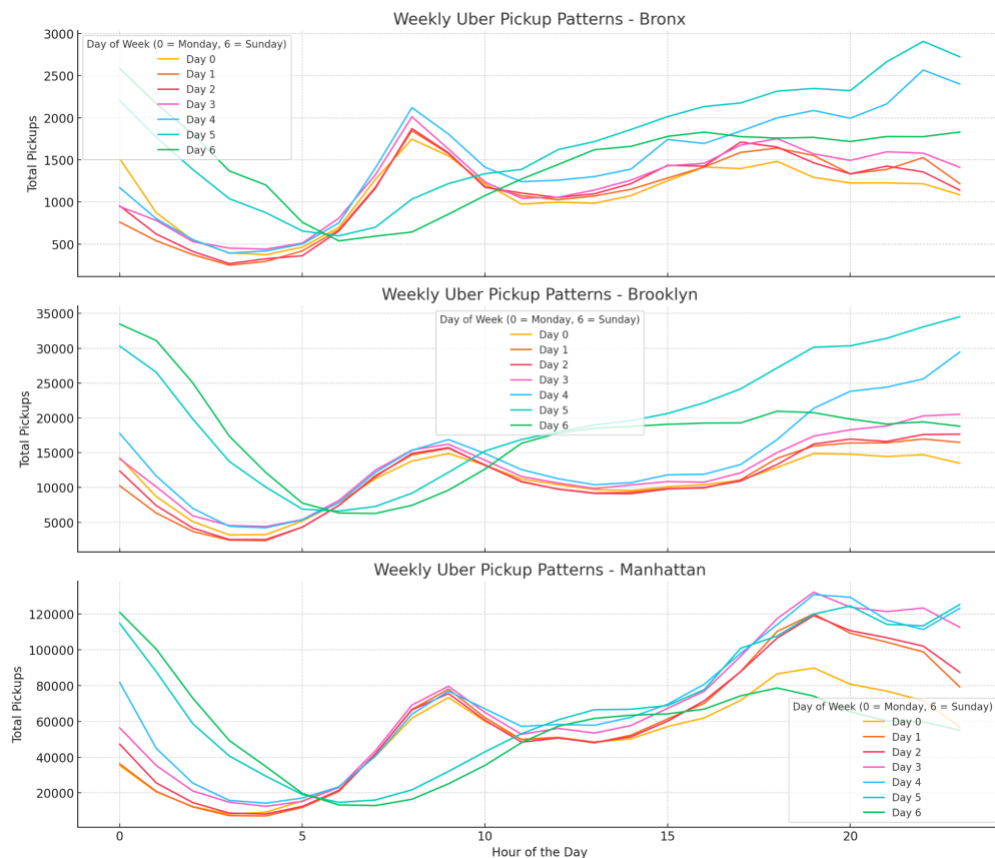


Figure 13: Average number of pickups by hour of the day for each day of the week, for three boroughs of New York City. Seasonal trends are very apparent in this view.

We found that in Manhattan, the peak of pickups occurred for the hours of 18:00, 19:00, 20:00, and 21:00, with over 700,000 pickups for these 4 hours in the 6 month period. During these periods, there was nearly double the demand than the average of all other hours of the day, indicating a high likelihood of surge pricing.

Brooklyn shared the same high pickup hours as Manhattan, from 18:00 through 21:00, showing around 60% higher demand during this period. This significant increase in demand also indicates a high likelihood of surge pricing.

The Bronx has a more even spread across the data, indicating the likelihood of surge pricing not being determined by pickup demand, but rather by driver availability. Its peak occurred between 17:00 and 22:00, but by less than half of the rate of increase as Manhattan. There was a noticeable increase in demand, but because it was not as pronounced as in Manhattan. This would indicate that these hours are the most likely to incur surge pricing as well.

What seasonal patterns appear in the data at the weekly level, and time of day level?

The data we analyzed showed significant seasonality at the weekly and time of day level. From our analysis, the below can be determined:

- Weekdays (Monday - Friday) during the morning rush hour (7:00 - 9:00) have high pickup rates, which correlates with travel to work in the morning.
- Evening Rush Hour (17:00 - 21:00), with the lowest increases happening on Sunday and Monday, and the highest on Thursday – Saturday.
- Very low pickup requests from 3:00 - 6:00 compared to all other times.
- Pickup request activity shifts to later in the day on weekends and extends into the late evening
- Saturday and Sunday consistently had the highest post-midnight activity.

Where should the most drivers be allocated to, and when, based on predictions of expected pickups?

Manhattan - Manhattan should have the highest density of drivers, with the most working during weekday mornings (7:00-9:00), evenings during rush hour (17:00 - 21:00) and late nights extending into the early next morning (22:00 - 2:00 the following day). Though we do not have precise data from our chart, it can be assumed that density during these periods in this borough should be where people are most likely to request from. For example, in the mornings, residential areas. For rush hour, the financial district or dining districts. And for the late nights, in the areas with nightlife.

Brooklyn - Brooklyn should have the highest density of drivers in residential areas for commuter traffic in the morning hours (7:00-9:00). Evenings should also have a higher density of drivers, focused on the dining districts and residential areas (18:00 - 21:00). Late evenings on the weekends (22:00 - 3:00 the following day) should also have a higher density of drivers.

Bronx - The Bronx showed a much more even distribution of ride requests. The only times where more drivers should be allocated were during the evening hours (17:00 - 22:00), likely near public transit hubs and residential areas, and early mornings (5:00 - 8:00), for commuter traffic to head into Manhattan or Brooklyn.

How does the number of pickups correlate with weather conditions and holiday schedule?

We have analyzed the correlation between the number of pickups and weather conditions in the exploratory data analysis section. We found that temperature and dew point have the highest correlation to the number of pickups. This indicates warmer temperatures may increase pickup requests. This likely also means that if it is hot, people are less likely to take transit or walk, and would prefer an Uber.

We also found a weak negative correlation between the number of pickups and precipitation and snow depth. This makes sense given that bad weather conditions result in less people wanting to travel or go out. We found through our analysis the correlation between uber pickups and holidays is -0.011, indicating a small negative correlation. This shows that there is really no impact on holidays and pickup requests.